

Write or stick your  
student number here



**2026** TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

# Mathematics Extension 1

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## General instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Student Number at the top of this page and at the top of each writing booklet

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**Total marks:**  
**70**

### **Section I – 10 marks (pages 2–6)**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

### **Section II – 60 marks (pages 8–12)**

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

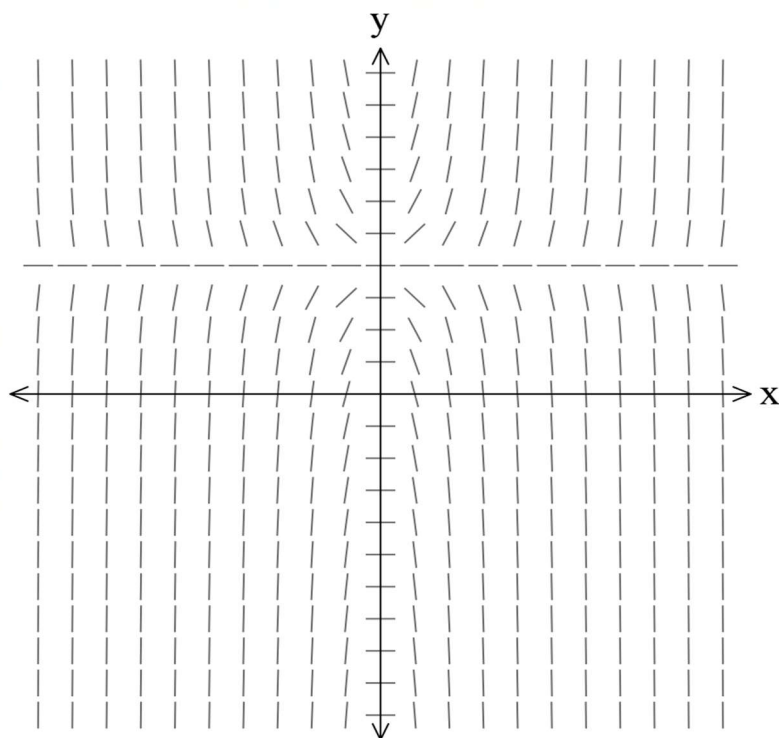
Use the multiple-choice answer sheet for Questions 1–10.

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- 1 Given the vectors  $\vec{a} = 2\vec{i} + 5\vec{j}$  and  $\vec{b} = -3\vec{i} + \vec{j}$ , what is the vector  $\vec{a} - \vec{b}$ ?
- A.  $-5\vec{i} - 4\vec{j}$
- B.  $-\vec{i} + 6\vec{j}$
- C.  $5\vec{i} + 4\vec{j}$
- D.  $5\vec{i} + 6\vec{j}$
- 2 What is the solution to the inequality  $2x^2 - x - 6 > 0$ ?
- A.  $-\frac{3}{2} < x < 2$
- B.  $x < -\frac{3}{2}$  or  $x > 2$
- C.  $x = -\frac{3}{2}$  or  $x = 2$
- D.  $-2 < x < \frac{3}{2}$
- 3 The expression  $\sqrt{3} \cos x - \sin x$  can be written in the form  $R \cos(x + \alpha)$ , where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$ . What are the values of  $R$  and  $\alpha$ ?
- A.  $R = 2, \alpha = \frac{11}{6}$
- B.  $R = 2, \alpha = \frac{\pi}{6}$
- C.  $R = 4, \alpha = \frac{\pi}{6}$
- D.  $R = 2, \alpha = \frac{\pi}{3}$

- 4 When the polynomial  $P(x) = x^3 + ax^2 - 5x + 6$  is divided by  $(x - 2)$ , the remainder is 12. What is the value of  $a$ ?
- A. 2
- B. -2
- C. 3
- D. 4

- 5 Consider the direction field below.



The direction field best represents the differential equation:

- A.  $\frac{dy}{dx} = \frac{x}{y} - 4x$
- B.  $\frac{dy}{dx} = y(y - 4)$
- C.  $\frac{dy}{dx} = \frac{x}{y-4}$
- D.  $\frac{dy}{dx} = xy - 4x$

- 6 What is the minimum number of students required in a class to guarantee that at least 6 students have their birthday in the same month?
- A. 61
- B. 72
- C. 66
- D. 60
- 7 A projectile is fired from the origin with an initial velocity of  $20\text{m/s}$  at an angle of  $30^\circ$  to the horizontal. Take the acceleration due to gravity as  $g = 10\text{m/s}^2$ .  
What is the maximum height reached by the projectile?
- A. 15 m
- B. 5 m
- C. 20 m
- D. 10 m

- 8 In the triangle  $OAB$ , the position vectors  $A$  and  $B$  relative to  $O$  are  $\underline{a}$  and  $\underline{b}$ , respectively, where

$$|\underline{a}| = 5, \quad |\underline{b}| = 7 \quad \text{and} \quad \underline{a} \cdot \underline{b} = 17.$$

The point  $P$  lies on the line  $AB$  such that  $OP$  is perpendicular to  $AB$ .

What is the length of  $OP$ ?

- A.  $\frac{3\sqrt{10}}{5}$
- B.  $\frac{3\sqrt{26}}{5}$
- C.  $\frac{3\sqrt{65}}{5}$
- D.  $\frac{6\sqrt{65}}{5}$
- 9 What is the value of the integral  $\int_0^3 \frac{1}{\sqrt{36-4x^2}} dx$ ?

- A.  $\frac{\pi}{4}$
- B.  $\frac{\pi}{6}$
- C.  $\frac{\pi}{12}$
- D.  $\frac{\pi}{2}$

- 10 The function  $f$  is defined by:

$$f(x) = \frac{2x - 3}{x + 2}, \quad x \neq -2.$$

The inverse function  $f^{-1}$  exists on its natural domain.

Which of the following is equal to  $f^{-1}(f(x + 1))$  for all values of  $x$  for which the expression is defined?

- A.  $x + 1$
- B.  $x + 1, \quad x \neq -3, x \neq 2$
- C.  $x + 1, \quad x \neq -3$
- D.  $x, \quad x \neq -3$

## Section II

**60 marks**

**Attempt Questions 11–14**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (17 marks) Use the Question 11 Writing Booklet

- a) Given the vectors  $\vec{u} = 2\vec{i} - 3\vec{j}$  and  $\vec{v} = -\vec{i} + 5\vec{j}$ , calculate the vector  $\text{proj}_{\vec{u}} \vec{v}$ . 2
- b) Solve  $\frac{2x-1}{x+3} \geq 1$  3
- c) Sketch the graph  $y = 3 \sin^{-1}(x)$ , labelling all key points. 2
- d) Evaluate the definite integral  $\int_0^4 \frac{x}{\sqrt{1+2x}} dx$  using the substitution  $u = 1 + 2x$ . 3
- e) The points  $A(1, -2)$ ,  $B(5, 1)$  and  $C(k, 2)$  form a triangle.
- i. Express  $\overrightarrow{AB}$  and  $\overrightarrow{AC}$  in component form. 1
- ii. Given  $\angle BAC = 90^\circ$ , find the value of  $k$ . 2
- iii. Hence, find the area of  $\triangle ABC$ . 1
- f) Use mathematical induction to prove that  $5^{2n} - 1$  is divisible by 24 for all positive integers  $n$ . 3

**End of Question 11**

**Question 12** (16 marks) Use the Question 12 Writing Booklet

- a) Given the graph of  $y = f(x)$  on page 1 of the Question 12 answer booklet, sketch  $y = \sqrt{f(x)}$  on the same number plane. **2**
- b) The function  $f(x) = x^2 + 1$  has domain  $x \geq 0$ . Find  $f^{-1}(10)$ . **1**
- c) In the expansion of  $(1 + 2x)^n$ , where  $n$  is a positive integer, the coefficient of  $x^3$  is eight times the coefficient of  $x^2$ .  
Find the value of  $n$ . **3**
- d) i. Prove that  $\cos 5x - \cos x = -2 \sin 2x \sin 3x$  **2**
- ii. Hence, solve  $\cos 5x - \cos x = \sin 3x$  for  $0 \leq x \leq \pi$ .  
Give all solutions in exact form. **3**
- e) Let  $\alpha, \beta$  and  $\gamma$  be the roots of the equation  $x^3 - 4x^2 + 2x + 3 = 0$
- i. Find the exact value of  $\alpha^2 + \beta^2 + \gamma^2$ . **2**
- ii. Find the exact value of  $\frac{1}{\alpha+\beta} + \frac{1}{\beta+\gamma} + \frac{1}{\gamma+\alpha}$  **3**

**End of Question 12**

**Question 13** (14 marks) Use the Question 13 Writing Booklet

a) Prove  $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} = 2$  **2**

- b) Eight people, including Chris, Josh and Dean, are seated around a circular table. **2**

Chris must sit directly opposite Josh. Dean must not sit next to Josh.

How many different seating arrangements are possible?

- c) A curve is defined parametrically by

$$x = 4 \sin \theta \quad \text{and} \quad y = 3 + 4 \cos \theta$$

- i. Show that the Cartesian equation of the curve is  $x^2 + (y - 3)^2 = 16$ . **2**

- ii. Show that  $\frac{dy}{dx} = -\tan \theta$  **1**

- iii. Find the equation of the tangent to the curve at the point corresponding to  $\theta = \frac{\pi}{4}$ . Give your answer in general form. **2**

- d) Consider the two functions  $f(x) = x^3 - 3x$  and  $g(x) = x^2 - x$ .

- i. Show that the curves intersect when  $x = -1, x = 0$  and  $x = 2$ . **2**

- ii. Find the exact total area enclosed between  $f(x)$  and  $g(x)$ . **3**

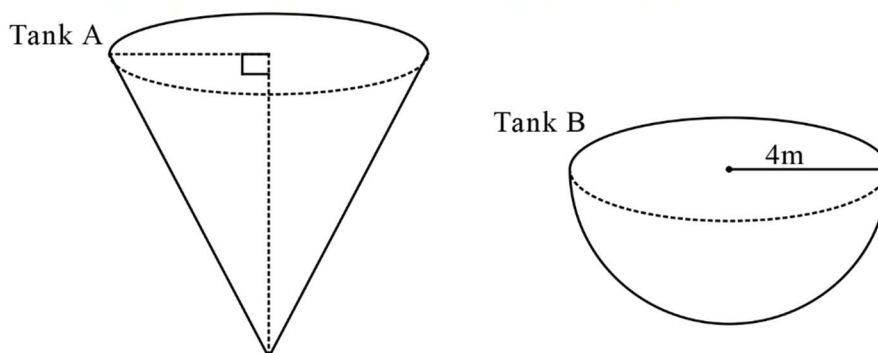
**End of Question 13**

**Question 14** (13 marks) Use the Question 14 Writing Booklet

- a) Two tanks contain a total of  $48\pi \text{ m}^3$  of water.

Tank A is in the shape of a right cone, as shown below. When the depth of water in Tank A is  $x$  m, the radius of the water surface is  $\frac{x}{2}$  m.

Tank B is hemispherical with a radius of 4 m. When the depth of water in Tank B is  $y$  m, the volume of water in Tank B is  $V_B = \pi y^2 \left(4 - \frac{y}{3}\right)$



Water is pumped from Tank A into Tank B at a constant rate of  $8\pi \text{ m}^3 \text{ min}^{-1}$ .

- |     |                                                                                                                                |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------|---|
| i.  | Find $\frac{dy}{dt}$                                                                                                           | 2 |
|     |                                                                                                                                |   |
| ii. | Find an expression for $t$ as a function of $y$ . Given that Tank B starts with water in the tank that is at a depth of 1.5 m. | 1 |

Question 14 continues on the next page

### Question 14 continued

- b) A function  $y$  satisfies

$$\frac{dy}{dx} = \frac{(y-1)(y+3)}{2x}$$

where  $x < 0$ , and  $y = 3$  when  $x = -1$ .

- i. Show that

1

$$\frac{1}{(y-1)(y+3)} = \frac{1}{4} \left( \frac{1}{y-1} - \frac{1}{y+3} \right)$$

- ii. Hence, find  $y$  as a function of  $x$ .

3

- iii. Determine the value of  $x$  for which  $y = 9$ .

1

- c) A projectile is launched from the origin with initial speed  $u \text{ m s}^{-1}$  at an angle  $\theta$  above the horizontal. It must pass through the point  $P(20, 15)$ , where distances are measured in metres. Take  $g = 10 \text{ m s}^{-2}$ .

- i. Show that  $\tan \theta$  must satisfy:

2

$$2000 \tan^2 \theta - 20u^2 \tan \theta + (2000 + 15u^2) = 0$$

- ii. By considering the discriminant of this quadratic equation, find the minimum possible value of  $u$ .

2

- iii. Find the corresponding launch angle, correct to the nearest degree.

1

**End of Question 14**

**End of paper**

Write or stick your  
student number here



**2026** TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

# Mathematics Extension 1

## **SOLUTIONS**

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### **General instructions**

- Reading time – 10 minutes
- Working time – 2 hours
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### **Total marks: 70**

#### **Section I – 10 marks (pages 2–6)**

- Attempt Questions 1–10
- Allow about 15 minutes for this section

#### **Section II – 60 marks (pages 8–12)**

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section



## Mathematics Extension 1

### Section I Multiple choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**       $2 + 4 =$       (A) 2      (B) 6      (C) 8      (D) 9  
                                         A ☐      B ☒      C ☐      D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒      B ☒      C ☐      D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒      B ☒ <sup>correct</sup>      C ☐      D ☐

### Section I: Student responses

- |     |   |                                  |   |                                  |   |                                  |   |                                  |
|-----|---|----------------------------------|---|----------------------------------|---|----------------------------------|---|----------------------------------|
| 1.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input checked="" type="radio"/> | D | <input type="radio"/>            |
| 2.  | A | <input type="radio"/>            | B | <input checked="" type="radio"/> | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 3.  | A | <input type="radio"/>            | B | <input checked="" type="radio"/> | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 4.  | A | <input checked="" type="radio"/> | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 5.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input checked="" type="radio"/> |
| 6.  | A | <input checked="" type="radio"/> | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 7.  | A | <input type="radio"/>            | B | <input checked="" type="radio"/> | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 8.  | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input checked="" type="radio"/> | D | <input type="radio"/>            |
| 9.  | A | <input checked="" type="radio"/> | B | <input type="radio"/>            | C | <input type="radio"/>            | D | <input type="radio"/>            |
| 10. | A | <input type="radio"/>            | B | <input type="radio"/>            | C | <input checked="" type="radio"/> | D | <input type="radio"/>            |

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

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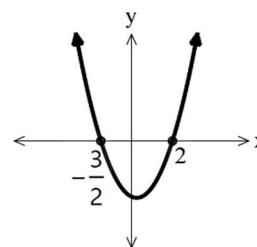
- 1 Given the vectors  $\vec{a} = 2\vec{i} + 5\vec{j}$  and  $\vec{b} = -3\vec{i} + \vec{j}$ , what is the vector  $\vec{a} - \vec{b}$ ?

$$\begin{aligned}\vec{a} - \vec{b} &= 2\vec{i} + 5\vec{j} - (-3\vec{i} + \vec{j}) \\ &= 5\vec{i} + 4\vec{j} \\ &\rightarrow C\end{aligned}$$

- 2 What is the solution to the inequality  $2x^2 - x - 6 > 0$ ?

$$(2x + 3)(x - 2) > 0$$

$$\begin{aligned}\therefore x &< -\frac{3}{2} \text{ or } x > 2 \\ &\rightarrow B\end{aligned}$$



- 3 The expression  $\sqrt{3} \cos x - \sin x$  can be written in the form  $R \cos(x + \alpha)$ , where  $R > 0$  and  $0 < \alpha < \frac{\pi}{2}$ . What are the values of  $R$  and  $\alpha$ ?

$$R \cos x \cos \alpha - R \sin x \sin \alpha = \sqrt{3} \cos x - \sin x$$

$$R = \sqrt{3 + 1}$$

$$= 2$$

$$\sin \alpha = 1 \quad \cos \alpha = \sqrt{3}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\rightarrow B$$

- 4 When the polynomial  $P(x) = x^3 + ax^2 - 5x + 6$  is divided by  $(x - 2)$ , the remainder is 12. What is the value of  $a$ ?

$$P(2) = 2^3 + a \times 2^2 - 5 \times 2 + 6 = 12$$

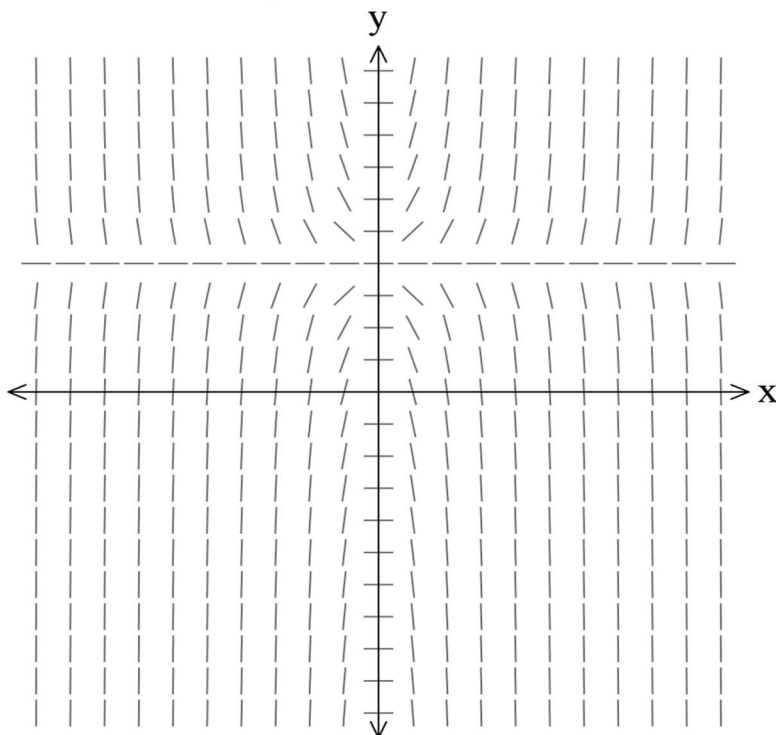
$$8 + 4a - 10 + 6 = 12$$

$$4a = 8$$

$$a = 2$$

$$\rightarrow A$$

- 5 Consider the direction field below.



The direction field best represents the differential equation:

$$y' = 0 \text{ when } x = 0 \rightarrow \text{not } B$$

$$y = 4 \text{ is a trivial solution} \rightarrow D$$

- 6 What is the minimum number of students required in a class to guarantee that at least 6 students have their birthday in the same month?

$$5 \times 12 + 1 = 61$$

→ A

- 7 A projectile is fired from the origin with an initial velocity of 20m/s at an angle of  $30^\circ$  to the horizontal. Take the acceleration due to gravity as  $g = 10\text{m/s}^2$ .

What is the maximum height reached by the projectile?

$$y'' = -10$$

$$y' = -10t + c$$

When  $t = 0, y' = 20 \sin 30 = 10 \text{ m/s}$

$$\therefore y' = -10t + 10$$

$$y = -5t^2 + 10t + c$$

When  $t = 0, y = 0$

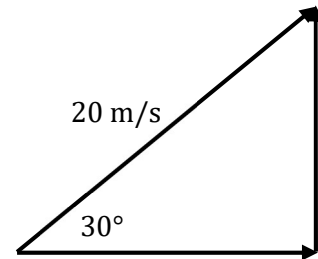
$$\therefore y = -5t^2 + 10t$$

Max at turning point  $t = -\frac{10}{-10} = 1$

$$y = -5(1)^2 + 10(1)$$

$$= 5 \text{ m}$$

→ B

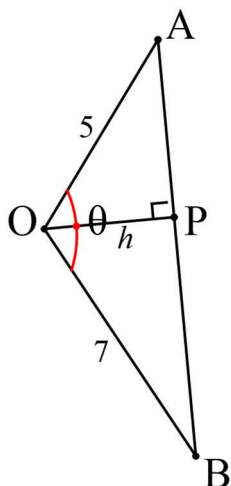


- 8 In the triangle  $OAB$ , the position vectors  $A$  and  $B$  relative to  $O$  are  $\underline{a}$  and  $\underline{b}$ , respectively, where

$$|\underline{a}| = 5, \quad |\underline{b}| = 7 \quad \text{and} \quad \underline{a} \cdot \underline{b} = 17.$$

The point  $P$  lies on the line  $AB$  such that  $OP$  is perpendicular to  $AB$ .

What is the length of  $OP$ ?



$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

$$17 = 5 \times 7 \times \cos \theta$$

$$\cos \theta = \frac{17}{35}$$

$$\theta = \cos^{-1} \frac{17}{35}$$

$$|\overrightarrow{AB}|^2 = 5^2 + 7^2 - 2 \times 5 \times 7 \times \frac{17}{35}$$

$$|\overrightarrow{AB}| = 2\sqrt{10}$$

$$\frac{\sin \theta}{2\sqrt{10}} = \frac{\sin A}{7}$$

$$\sin A = \frac{7 \times \sin \left( \cos^{-1} \frac{17}{35} \right)}{2\sqrt{10}}$$

$$\cos^{-1} \frac{17}{35} = \alpha \rightarrow \cos \alpha = \frac{17}{35}$$

$$b = \sqrt{35^2 - 17^2}$$

$$b = 6\sqrt{26}$$

$$\frac{7}{2\sqrt{10}} \times \frac{6\sqrt{26}}{35} = \sin A$$

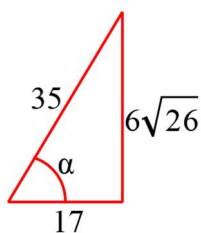
$$\sin A = \frac{3\sqrt{65}}{25}$$

$$\sin A = \frac{h}{5}$$

$$h = 5 \times \frac{3\sqrt{65}}{25}$$

$$|\overrightarrow{OP}| = \frac{3\sqrt{65}}{5}$$

$\rightarrow C$



9 What is the value of the integral  $\int_0^3 \frac{1}{\sqrt{36-4x^2}} dx$ ?

$$\begin{aligned} &= \frac{1}{2} \int_0^3 \frac{1}{\sqrt{9-x^2}} dx \\ &= \frac{1}{2} \left[ \sin^{-1} \left( \frac{x}{3} \right) \right]_0^3 \\ &= \frac{1}{2} (\sin^{-1}(1) - \sin^{-1}(0)) \\ &= \frac{1}{2} \left( \frac{\pi}{2} \right) \\ &= \frac{\pi}{4} \\ &\rightarrow A \end{aligned}$$

10 The function  $f$  is defined by:

$$f(x) = \frac{2x - 3}{x + 2}, \quad x \neq -2.$$

The inverse function  $f^{-1}$  exists on its natural domain.

Which of the following is equal to  $f^{-1}(f(x + 1))$  for all values of  $x$  for which the expression is defined?

$$\begin{aligned} f(x + 1) &= \frac{2(x + 1) - 3}{x + 1 + 2} \\ &= \frac{2x - 1}{x + 3} \\ x &\neq -3 \end{aligned}$$

Finding the inverse of  $f(x)$ :  $x = \frac{2y-3}{y+2}$

$$xy + 2x = 2y - 3$$

$$y(2 - x) = 2x + 3$$

$$f^{-1}(x) = \frac{2x + 3}{2 - x}$$

$$f^{-1}(f(x + 1)) = \frac{2\left(\frac{2x - 1}{x + 3}\right) + 3}{2 - \left(\frac{2x - 1}{x + 3}\right)}$$

$$= \frac{2(2x - 1) + 3(x + 3)}{2(x + 3) - (2x - 1)}$$

$$= \frac{4x - 2 + 3x + 9}{2x + 6 - 2x + 1}$$

$$= \frac{7x + 7}{7}$$

$$= x + 1$$

$$\rightarrow C$$

## Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

---

**Question 11** (17 marks) Use the Question 11 Writing Booklet

- a) Given the vectors  $\vec{u} = 2\vec{i} - 3\vec{j}$  and  $\vec{v} = -\vec{i} + 5\vec{j}$ , calculate the vector  $\text{proj}_{\vec{u}} \vec{v}$ . 2

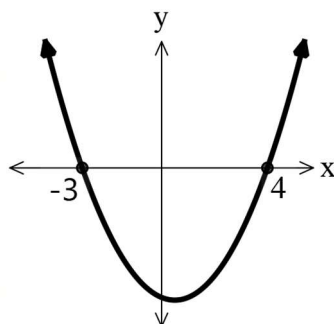
$$\begin{aligned}\text{proj}_{\vec{u}} \vec{v} &= \frac{\vec{u} \cdot \vec{v}}{|\vec{u}|^2} \vec{u} \\&= \frac{-2 - 15}{(\sqrt{2^2 + (-3)^2})^2} \begin{bmatrix} 2 \\ -3 \end{bmatrix} \\&= \frac{-17}{13} \begin{bmatrix} 2 \\ -3 \end{bmatrix} \\&= -\frac{34}{13} \vec{i} + \frac{51}{13} \vec{j}\end{aligned}$$

| Marks | Criteria                                                                                                                                    |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | Correct solution                                                                                                                            |
| 1     | Progress towards solution such as finding the projection of $\vec{u}$ onto $\vec{v}$ or correctly substituting into the projection formula. |

b) Solve  $\frac{2x-1}{x+3} \geq 1$

3

$$\begin{aligned} x &\neq -3 \\ (2x-1)(x+3) &\geq (x+3)^2 \\ (x+3)[2x-1-(x+3)] &\geq 0 \\ (x+3)(x-4) &\geq 0 \end{aligned}$$

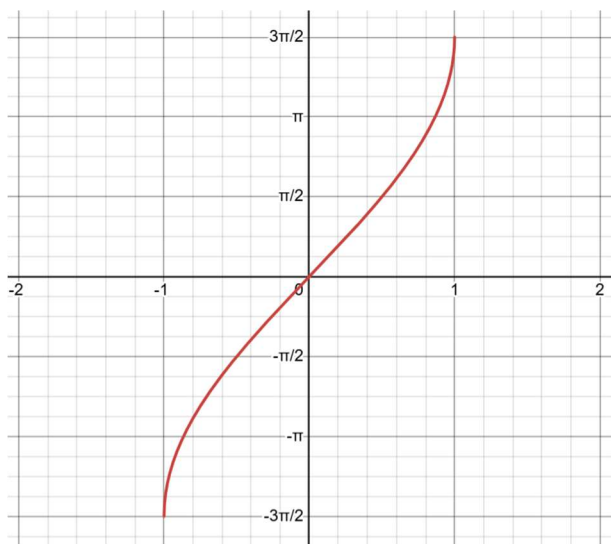


$$\therefore x < -3, x \geq 4 \text{ since } x \neq -3$$

| Marks | Criteria                                                                                                                                                                   |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3     | Correct solution with testing the appropriate regions.                                                                                                                     |
| 2     | Allow 2 marks for finding the critical points <b>and</b> testing regions either graphically or through calculations but incorrect inequalities are given for the solution. |
| 1     | Allow 1 mark for finding the critical points or identifying regions in some way.                                                                                           |

c) Sketch the graph  $y = 3 \sin^{-1}(x)$ , labelling all key points.

2



| Marks | Criteria                                                                                                  |
|-------|-----------------------------------------------------------------------------------------------------------|
| 2     | Correct graph. Clear endpoints at $\left(-1, -\frac{3\pi}{2}\right)$ and $\left(1, \frac{3\pi}{2}\right)$ |
| 1     | One error such as incorrect dilation or making the graph continue.                                        |

d) Evaluate the definite integral  $\int_0^4 \frac{x}{\sqrt{1+2x}} dx$  using the substitution  $u = 1 + 2x$ . 3

$$\frac{du}{dx} = 2$$

$$\frac{1}{2} du = dx$$

$$\text{When } x = 0, u = 1$$

$$\text{When } x = 4, u = 9$$

$$x = \frac{1}{2}(u - 1)$$

$$\therefore \int_0^4 \frac{x}{\sqrt{1+2x}} dx = \int_1^9 \frac{1}{2} \frac{u-1}{\sqrt{u}} \left(\frac{1}{2}\right) du$$

$$= \frac{1}{4} \int_1^9 u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$$

$$= \frac{1}{4} \left[ \frac{u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{u^{\frac{1}{2}}}{\frac{1}{2}} \right]_1^9$$

$$= \frac{1}{4} \left[ \frac{2\sqrt{9^3}}{3} - 2\sqrt{9} - \left( \frac{2}{3} - 2 \right) \right]$$

$$= \frac{10}{3}$$

| Marks | Criteria                                                                                          |
|-------|---------------------------------------------------------------------------------------------------|
| 3     | Correct solution                                                                                  |
| 2     | Correct solution but one mistake made such as not changing the bounds or making a numerical error |
| 1     | Correctly makes a substitution for $dx$ , $x$ , or the bounds.                                    |

e) The points  $A(1, -2)$ ,  $B(5, 1)$  and  $C(k, 2)$  form a triangle.

- i. Express  $\overrightarrow{AB}$  and  $\overrightarrow{AC}$  in component form.

1

$$\overrightarrow{AB} = 4\underset{\sim}{i} + 3\underset{\sim}{j}$$

$$\overrightarrow{AC} = (k-1)\underset{\sim}{i} + 4\underset{\sim}{j}$$

| Marks                                       | Criteria                                              |
|---------------------------------------------|-------------------------------------------------------|
| 1                                           | Both correct. Accept component or column vector form. |
| <b>Comment:</b><br>Well done by the cohort. |                                                       |

- ii. Given  $\angle BAC = 90^\circ$ , find the value of  $k$ .

2

$$\overrightarrow{AB} \cdot \overrightarrow{AC} = 4(k-1) + 3 \times 4 = 0$$

$$4k - 4 + 12 = 0$$

$$4k = -8$$

$$k = -2$$

| Marks                                                                                               | Criteria                                        |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 2                                                                                                   | Finds the correct solution for $k$              |
| 1                                                                                                   | Acknowledges that the dot product must equal 0. |
| <b>Comment:</b><br>Mostly well done. Great to see so many students knowing their vector properties. |                                                 |

- iii. Hence, find the area of  $\triangle ABC$ .

1

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} \times |\overrightarrow{AB}| |\overrightarrow{AC}| \\
 &= \frac{1}{2} \times \sqrt{3^2 + 4^2} \times \sqrt{(-3)^2 + 4^2} \\
 &= \frac{1}{2} \times 5 \times 5 \\
 &= 12.5 \text{ u}^2
 \end{aligned}$$

| Marks                                                                                                                                    | Criteria        |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1                                                                                                                                        | Correct answer. |
| <b>Comment:</b><br>Well done. Carried errors were given base off the answer to (ii). The magnitude of AC comes from your answer to (ii). |                 |

- f) Use mathematical induction to prove that  $5^{2n} - 1$  is divisible by 24 for all positive integers  $n$ .

3

Prove true for  $n = 1$

$$5^{2n} - 1 = 5^{2 \times 1} - 1$$

$$= 25 - 1$$

$$= 24 \text{ (which is divisible by 24)}$$

$\therefore$  true for  $n = 1$

Assume true for  $n = k, k \in \mathbb{Z}$ :

$$5^{2k} - 1 = 24M, M \in \mathbb{Z}$$

Prove true for  $n = k + 1$ , if true for  $n = k$

$$\text{R.t.p } 5^{2(k+1)} - 1 = 24P, P \in \mathbb{Z}$$

$$\text{LHS} = 5^{2k+2} - 1$$

$$= 5^2 \times 5^{2k} - 1$$

$$= 25 \times (24M + 1) - 1 \text{ from assumption}$$

$$= 25 \times 24M + 25 - 1$$

$$= 24(25M + 1)$$

$$= 24P \text{ where } P = 25M + 1 \in \mathbb{Z}$$

$\therefore$  true for  $n = k + 1$  if true for  $n = k$

Since true for  $n = 1$ , true for all  $n \geq 1$

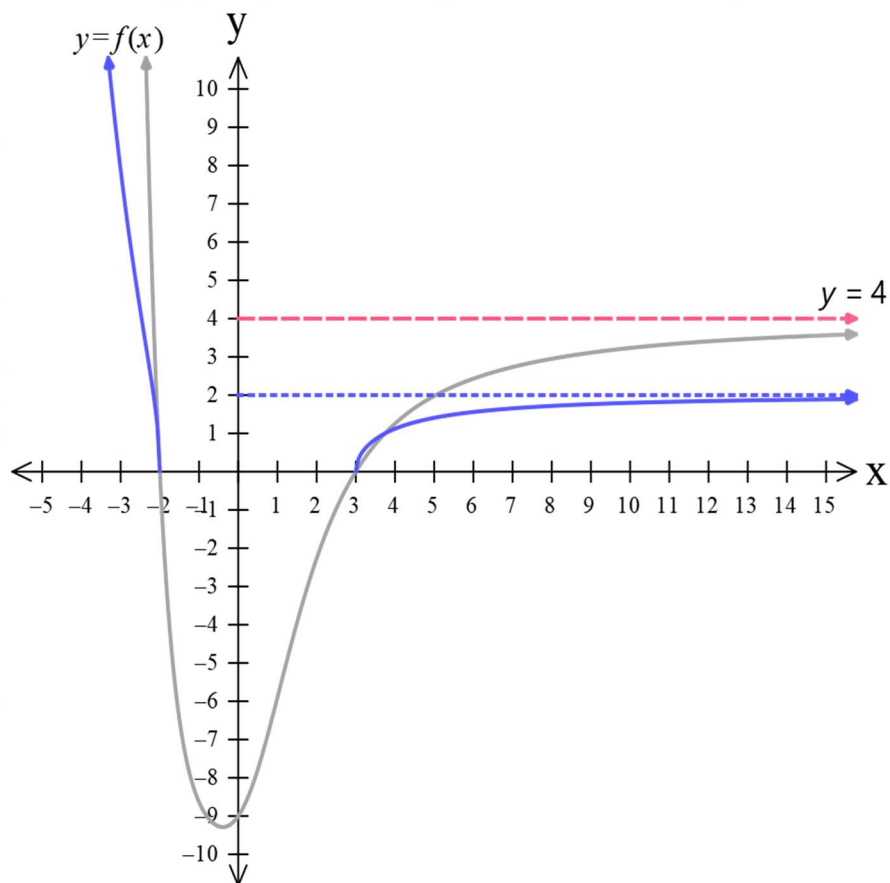
| Marks | Criteria                                                                                    |
|-------|---------------------------------------------------------------------------------------------|
| 3     | Correct solution with all required elements and all introduced variables defined correctly. |
| 2     | Correct base case, assumption step and an attempt at starting the inductive step            |
| 1     | Correct base case                                                                           |

**End of Question 11**

**Question 12** (16 marks) Use the Question 12 Writing Booklet

- a) Given the graph of  $y = f(x)$  on page 1 of the Question 12 answer booklet, sketch  $y = \sqrt{f(x)}$  on the same number plane.

2



| Marks | Criteria                                                                                  |
|-------|-------------------------------------------------------------------------------------------|
| 2     | Correct graph, with new labelled asymptote and clear intersections points when $f(x) = 1$ |
| 1     | Some correct elements but not all necessary points shown.                                 |

- b) The function  $f(x) = x^2 + 1$  has domain  $x \geq 0$ . Find  $f^{-1}(10)$ .  
To find  $f^{-1}(10)$ , find when  $f(x) = 10$ .

1

$$\begin{aligned}
 10 &= x^2 + 1 \\
 x^2 &= 9 \\
 x &= 3 \text{ since } x \geq 0 \\
 \therefore f^{-1}(10) &= 3
 \end{aligned}$$

| Marks | Criteria        |
|-------|-----------------|
| 1     | Correct answer. |

- c) In the expansion of  $(1 + 2x)^n$ , where  $n$  is a positive integer, the coefficient of  $x^3$  is eight times the coefficient of  $x^2$ . 3  
Find the value of  $n$ .

$$(1 + 2x)^n = {}^nC_0 + {}^nC_1 2x + {}^nC_2 (2x)^2 + {}^nC_3 (2x)^3 \dots$$

$$8 \times {}^nC_2 \times 2^2 = {}^nC_3 \times 2^3$$

$$32 \times \frac{n!}{(n-2)! 2!} = \frac{n!}{(n-3)! 3!} \times 8$$

$$\frac{16}{(n-2)!} = \frac{4}{3(n-3)!}$$

$$48(n-3)! = 4(n-2)!$$

$$48 = 4(n-2)$$

$$4n = 56$$

$$n = 14$$

| Marks | Criteria                                                                                                  |
|-------|-----------------------------------------------------------------------------------------------------------|
| 3     | Correct solution.                                                                                         |
| 2     | Writes an equation comparing coefficients between the $x^2$ and $x^3$ terms.                              |
| 1     | Finds an expression for the $n$ th term of the expansion or demonstrates the expansion to the $x^3$ term. |

- d) i. Prove that  $\cos 5x - \cos x = -2 \sin 2x \sin 3x$  2

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\cos(A - B) - \cos(A + B) = 2 \sin A \sin B$$

$$\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$$

$$\text{Let } A = 3x \text{ and } B = 2x$$

$$\text{Then } \cos 5x - \cos x = -2 \sin 3x \sin 2x$$

| Marks | Criteria                         |
|-------|----------------------------------|
| 2     | Correct proof with logical steps |
| 1     | Attempts to use sums to products |

- ii. Hence, solve  $\cos 5x - \cos x = \sin 3x$  for  $0 \leq x \leq \pi$ . 3  
Give all solutions in exact form.

$$-2 \sin 2x \sin 3x = \sin 3x \quad 0 \leq 2x \leq 2\pi, \quad 0 \leq 3x \leq 3\pi$$

$$\sin 3x + 2 \sin 2x \sin 3x = 0$$

$$\sin 3x (1 + 2 \sin 2x) = 0$$

$$\sin 3x = 0 \quad \text{or} \quad \sin 2x = -\frac{1}{2}$$

$$3x = 0, \pi, 2\pi, 3\pi \quad \text{or} \quad 2x = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore x = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{7\pi}{12}, \frac{11\pi}{12}$$

| Marks | Criteria                                                                                 |
|-------|------------------------------------------------------------------------------------------|
| 3     | All correct answers given with appropriate working.                                      |
| 2     | All correct solutions from one factor OR some correct solutions from each of the factors |
| 1     | At least 1 correct solution from correct working                                         |

e) Let  $\alpha, \beta$  and  $\gamma$  be the roots of the equation  $x^3 - 4x^2 + 2x + 3 = 0$

i. Find the exact value of  $\alpha^2 + \beta^2 + \gamma^2$ . 2

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) \\ &= 4^2 - 2(2) \\ &= 12\end{aligned}$$

| Marks | Criteria                                                                                                   |
|-------|------------------------------------------------------------------------------------------------------------|
| 2     | Correct expression and value.                                                                              |
| 1     | Attempts to expand $(\alpha + \beta + \gamma)^2$ to find an expression for $\alpha^2 + \beta^2 + \gamma^2$ |

ii. Find the exact value of  $\frac{1}{\alpha+\beta} + \frac{1}{\beta+\gamma} + \frac{1}{\gamma+\alpha}$  3

$$\alpha + \beta + \gamma = 4$$

$$\begin{aligned}\frac{1}{\alpha + \beta} + \frac{1}{\beta + \gamma} + \frac{1}{\gamma + \alpha} &= \frac{1}{4 - \gamma} + \frac{1}{4 - \alpha} + \frac{1}{4 - \beta} \\ &= \frac{(4 - \alpha)(4 - \beta) + (4 - \gamma)(4 - \beta) + (4 - \gamma)(4 - \alpha)}{(4 - \gamma)(4 - \beta)(4 - \alpha)} \\ &= \frac{16 - 4\alpha - 4\beta + \alpha\beta + 16 - 4\beta - 4\gamma + \beta\gamma + 16 - 4\alpha - 4\gamma + \alpha\gamma}{(4 - \gamma)(16 - 4\alpha - 4\beta + \alpha\beta)} \\ &= \frac{48 - 8(\alpha + \beta + \gamma) + (\alpha\beta + \beta\gamma + \alpha\gamma)}{64 - 16\alpha - 16\beta + 4\alpha\beta - 16\gamma + 4\alpha\gamma + 4\beta\gamma - \alpha\beta\gamma} \\ &= \frac{48 - 8(\alpha + \beta + \gamma) + (\alpha\beta + \beta\gamma + \alpha\gamma)}{64 - 16(\alpha + \beta + \gamma) + 4(\alpha\beta + \beta\gamma + \alpha\gamma) - \alpha\beta\gamma} \\ &= \frac{48 - 8(4) + 2}{64 - 16(4) + 4(2) - (-3)} \\ &= \frac{18}{11}\end{aligned}$$

| Marks | Criteria                                                                                                        |
|-------|-----------------------------------------------------------------------------------------------------------------|
| 3     | Correct solution                                                                                                |
| 2     | Correctly forms a single fraction and finds the correct value for numerator or denominator with correct working |
| 1     | Makes the expression into a correct single fraction                                                             |

**End of Question 12**

**Question 13** (14 marks) Use the Question 13 Writing Booklet

a) Prove  $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} = 2$

2

$$\begin{aligned} LHS &= \frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} \\ &= \frac{\sin 3x \cos x - \cos 3x \sin x}{\sin x \cos x} \\ &= \frac{\sin(3x - x)}{\sin x \cos x} \\ &= \frac{\sin 2x}{\frac{1}{2} \sin 2x} \\ &= \frac{\sin 2x}{\frac{1}{2} \sin 2x} \\ &= 2 \\ &= RHS \end{aligned}$$

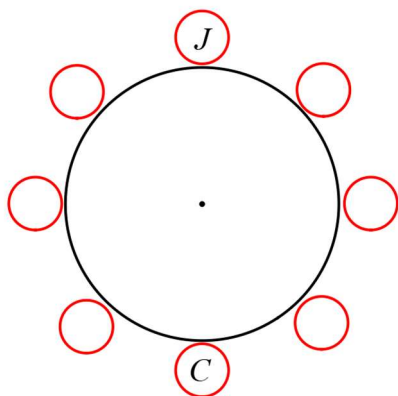
| Marks | Criteria                                         |
|-------|--------------------------------------------------|
| 2     | Correct proof with logical steps.                |
| 1     | Uses products to sums or compound angle formulas |

b) Eight people, including Chris, Josh and Dean, are seated around a circular table.

2

Chris must sit directly opposite Josh. Dean must not sit next to Josh.

How many different seating arrangements are possible?



Seat Josh first. Only 1 option for Chris. 4 options for D. 5! Ways to seat others.

$$\text{Total} = 4 \times 5!$$

$$= 480$$

Typical calculations shown by student that showed valid engagement with this problem were:

$5 \times 4 \times 4!$  (Seating Josh and Chris,  $5 \times 4$  ways of seating people either side of Josh,  $4!$  ways of seating the remaining people)

$6! - 2 \times 5!$  (Seating Josh and Chris,  $6!$  ways of arranging remaining people, remove the arrangements where Dean sits next to Josh)

$(7-1)! \times \frac{4}{6}$  (Consider Josh and Chris to be one group, 7 groups arranged

around a circle, there is a  $\frac{4}{6}$  chance that Dean will not be seated next to Josh.

| Marks | Criteria                                                                                                                                                                                                                                                                                                                                                                         |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | Correct answer.                                                                                                                                                                                                                                                                                                                                                                  |
| 1     | Makes at least 1 logical conclusion but error in values selected such as not accounting for rotations.<br>Students who received 1 mark typically had answers of $2 \times 480$ as they rearranged Chris and Josh, but this created repeated arrangements or $4! 5!$<br>Students who wrote $6!$ Received 0 marks as there was insufficient knowledge of the problem demonstrated. |

c) A curve is defined parametrically by

$$x = 4 \sin \theta \quad \text{and} \quad y = 3 + 4 \cos \theta$$

i. Show that the Cartesian equation of the curve is  $x^2 + (y - 3)^2 = 16$ . 2

$$\sin \theta = \frac{x}{4} \quad \text{and} \quad \cos \theta = \frac{y - 3}{4}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\left(\frac{x}{4}\right)^2 + \left(\frac{y - 3}{4}\right)^2 = 1$$

$$\frac{x^2}{16} + \frac{(y - 3)^2}{16} = 1$$

$$x^2 + (y - 3)^2 = 16$$

| Marks | Criteria                                   |
|-------|--------------------------------------------|
| 2     | Correct solution with appropriate working. |
| 1     | Attempts to use the Pythagorean Identity.  |

ii. Show that  $\frac{dy}{dx} = -\tan \theta$  1

$$y = 3 + 4 \cos \theta$$

$$\frac{dy}{d\theta} = -4 \sin \theta$$

$$x = 4 \sin \theta$$

$$\frac{dx}{d\theta} = 4 \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$\begin{aligned}
 &= -4 \sin \theta \times \frac{1}{4 \cos \theta} \\
 &= -\tan \theta
 \end{aligned}$$

Another valid approach attempted is below, but students were generally awarded 0 marks due to small, but significant details missing.

$$\begin{aligned}
 x &= 4 \sin \theta \quad \textcircled{1} \\
 y &= 3 + 4 \cos \theta & \Rightarrow y - 3 = 4 \cos \theta \\
 x^2 + (y - 3)^2 &= 16 \\
 y - 3 &= \pm \sqrt{16 - x^2} & \Rightarrow \pm \sqrt{16 - x^2} = 4 \cos \theta \quad \textcircled{2} \\
 y &= 3 \pm \sqrt{16 - x^2} \\
 \frac{dy}{dx} &= \left( \pm \frac{1}{2} (16 - x^2)^{-\frac{1}{2}} \times -2x \right) \\
 &= \frac{-x}{\pm \sqrt{16 - x^2}} \\
 &= \frac{-4 \sin \theta}{4 \cos \theta} \text{ from } \textcircled{1} \text{ and } \textcircled{2} \\
 &= -\tan \theta \text{ as required.}
 \end{aligned}$$

| Marks | Criteria                              |
|-------|---------------------------------------|
| 1     | Correct solution using the chain rule |

- iii. Find the equation of the tangent to the curve at the point corresponding to  $\theta = \frac{\pi}{4}$ . Give your answer in general form. 2

$$\begin{aligned}
 x &= 4 \sin \frac{\pi}{4} \\
 &= 4 \left( \frac{1}{\sqrt{2}} \right) \\
 &= 2\sqrt{2} \\
 y &= 3 + 4 \cos \frac{\pi}{4} \\
 &= 3 + 4 \left( \frac{1}{\sqrt{2}} \right) \\
 &= 3 + 2\sqrt{2} \\
 m &= -\tan \frac{\pi}{4} \\
 &= -1 \\
 y - y_1 &= m(x - x_1) \\
 y - (3 + 2\sqrt{2}) &= -1(x - 2\sqrt{2}) \\
 x + y - 3 - 4\sqrt{2} &= 0
 \end{aligned}$$

| Marks | Criteria                                                                                                        |
|-------|-----------------------------------------------------------------------------------------------------------------|
| 2     | Correct equation in any appropriate form.                                                                       |
| 1     | Correctly finds the gradient or the point <b>and</b> attempts to find the equation using an appropriate method. |

- d) Consider the two functions  $f(x) = x^3 - 3x$  and  $g(x) = x^2 - x$ .  
 i. Show that the curves intersect when  $x = -1, x = 0$  and  $x = 2$ .

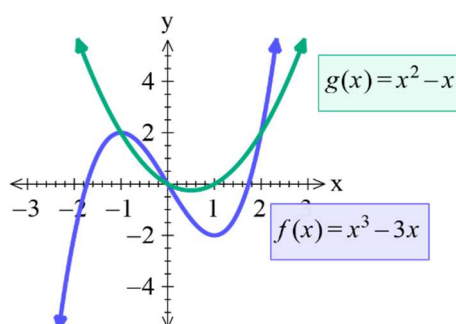
2

$$\begin{aligned}x^3 - 3x &= x^2 - x \\x^3 - x^2 - 2x &= 0 \\x(x^2 - x - 2) &= 0 \\x(x - 2)(x + 1) &= 0 \\\therefore x &= 0, 2, -1\end{aligned}$$

| Marks | Criteria                                    |
|-------|---------------------------------------------|
| 2     | Correct solution and showing logical steps. |
| 1     | Equates the two functions, or               |

- ii. Find the exact total area enclosed between  $f(x)$  and  $g(x)$ .

3



$$\begin{aligned}A &= \int_{-1}^0 (x^3 - 3x) - (x^2 - x) \, dx + \int_0^2 (x^2 - x) - (x^3 - 3x) \, dx \\&= \int_{-1}^0 x^3 - x^2 - 2x \, dx + \int_0^2 2x + x^2 - x^3 \, dx \\&= \left[ \frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_{-1}^0 + \left[ x^2 + \frac{x^3}{3} - \frac{x^4}{4} \right]_0^2 \\&= 0 - \left( \frac{1}{4} + \frac{1}{3} - 1 \right) + \left( 4 + \frac{8}{3} - 4 \right) - 0 \\&= \frac{37}{12} \text{ u}^2\end{aligned}$$

| Marks | Criteria                                                                                                                     |
|-------|------------------------------------------------------------------------------------------------------------------------------|
| 3     | Correct solution                                                                                                             |
| 2     | Significant progress such as setting up 2 distinct integrals for the 2 areas and attempting to solve them with small errors. |
| 1     | Solves an integral involving the difference of the two functions                                                             |

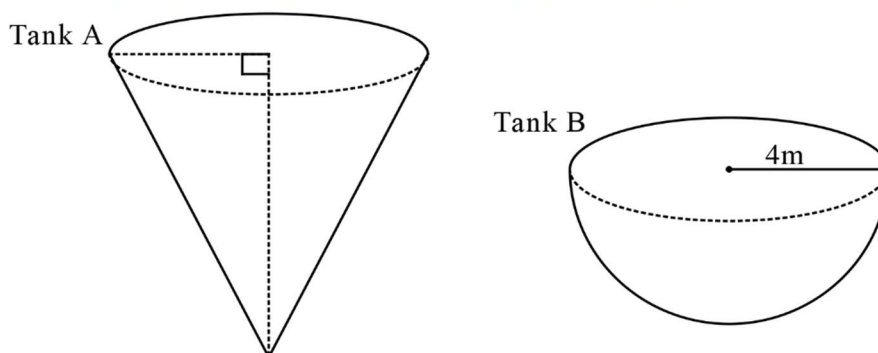
End of Question 13

**Question 14** (13 marks) Use the Question 14 Writing Booklet

- a) Two tanks contain a total of  $48\pi \text{ m}^3$  of water.

Tank A is in the shape of a right cone, as shown below. When the depth of water in Tank A is  $x$  m, the radius of the water surface is  $\frac{x}{2}$  m.

Tank B is hemispherical with a radius of 4 m. When the depth of water in Tank B is  $y$  m, the volume of water in Tank B is  $V_B = \pi y^2 \left(4 - \frac{y}{3}\right)$



Water is pumped from Tank A into Tank B at a constant rate of  $8\pi \text{ m}^3 \text{ min}^{-1}$ .

- i. Find  $\frac{dy}{dt}$

2

$$\frac{dV_B}{dy} = 8\pi y - \pi y^2$$

$$\frac{dV_B}{dt} = 8\pi$$

$$\frac{dy}{dt} = \frac{dV_B}{dt} \times \frac{dy}{dV_B}$$

$$= 8\pi \times \frac{1}{\pi(8y - y^2)}$$

$$= \frac{8}{8y - y^2}$$

| Marks | Criteria                                                                                          |
|-------|---------------------------------------------------------------------------------------------------|
| 2     | Correct solution.                                                                                 |
| 1     | Correctly find $\frac{dV_B}{dy}$ or uses chain rule correctly with an incorrect $\frac{dV_B}{dy}$ |

- ii. Find an expression for  $t$  as a function of  $y$ . Given that Tank B starts with water in the tank that is at a depth of 1.5 m. 1

$$\frac{dy}{dt} = \frac{8}{8y - y^2}$$

$$\int 8y - y^2 dy = \int 8 dt$$

$$4y^2 - \frac{y^3}{3} = 8t + C$$

When  $t = 0, y = 1.5$

$$4 \times 1.5^2 - \frac{1.5^3}{3} = C$$

$$C = \frac{63}{8}$$

$$8t = 4y^2 - \frac{y^3}{3} - \frac{63}{8}$$

$$t = \frac{y^2}{2} - \frac{y^3}{24} - \frac{63}{64}$$

$$t = \frac{96y^2 - 8y^3 - 189}{192}$$

| Marks | Criteria          |
|-------|-------------------|
| 1     | Correct solution. |

### Question 14 continued

- b) A function  $y$  satisfies

$$\frac{dy}{dx} = \frac{(y-1)(y+3)}{2x}$$

where  $x < 0$ , and  $y = 3$  when  $x = -1$ .

- i. Show that

$$\frac{1}{(y-1)(y+3)} = \frac{1}{4} \left( \frac{1}{y-1} - \frac{1}{y+3} \right)$$

1

$$\begin{aligned}
 RHS &= \frac{1}{4} \left( \frac{1}{y-1} - \frac{1}{y+3} \right) \\
 &= \frac{1}{4} \left( \frac{y+3 - (y-1)}{(y-1)(y+3)} \right) \\
 &= \frac{1}{4} \left( \frac{4}{(y-1)(y+3)} \right) \\
 &= \frac{1}{(y-1)(y+3)} \\
 &= LHS
 \end{aligned}$$

| Marks | Criteria         |
|-------|------------------|
| 1     | Correct solution |

ii. Hence, find  $y$  as a function of  $x$ .

3

$$\frac{dy}{dx} = \frac{(y-1)(y+3)}{2x}$$

$$\int \frac{1}{(y-1)(y+3)} dy = \int \frac{1}{2x} dx$$

$$\frac{1}{4} \int \frac{1}{y-1} - \frac{1}{y+3} dy = \frac{1}{2} \int \frac{1}{x} dx$$

$$\frac{1}{2} (\ln|y-1| - \ln|y+3|) = \ln|x| + C$$

$$\frac{1}{2} \left( \ln \left| \frac{y-1}{y+3} \right| \right) = \ln|x| + C$$

$$\ln \left| \frac{y-1}{y+3} \right| = 2 \ln|x| + C_2$$

$$\left| \frac{y-1}{y+3} \right| = e^{2 \ln(-x) + C_2} \text{ since } x < 0$$

$$\frac{y-1}{y+3} = A e^{\ln^2} \text{ where } A = \pm e^{C_2}$$

$$\frac{y-1}{y+3} = A x^2$$

When  $y = 3, x = -1$

$$\frac{3-1}{3+3} = A(-1)^2$$

$$\frac{1}{3} = A$$

$$\therefore \frac{3y-3}{y+3} = x^2$$

$$3y-3 = x^2y + 3x^2$$

$$3y - x^2y = 3x^2 + 3$$

$$y(3 - x^2) = 3(x^2 + 1)$$

$$y = \frac{3(x^2 + 1)}{3 - x^2}$$

| Marks | Criteria                                         |
|-------|--------------------------------------------------|
| 3     | Correct solution                                 |
| 2     | Finds the value of A                             |
| 1     | Separates the integral and substitutes in part i |

iii. Determine the value of  $x$  for which  $y = 9$ .

1

$$9 = \frac{3(x^2 + 1)}{3 - x^2}$$

$$27 - 9x^2 = 3x^2 + 3$$

$$12x^2 = 24$$

$$x^2 = 2$$

$$x = -\sqrt{2}$$

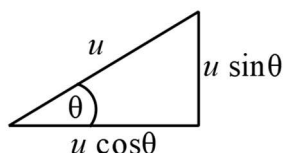
| Marks | Criteria       |
|-------|----------------|
| 1     | Correct answer |

- c) A projectile is launched from the origin with initial speed  $u \text{ m s}^{-1}$  at an angle  $\theta$  above the horizontal. It must pass through the point  $P(20, 15)$ , where distances are measured in metres. Take  $g = 10 \text{ m s}^{-2}$ .

- i. Show that  $\tan \theta$  must satisfy:

2

$$2000 \tan^2 \theta - 20u^2 \tan \theta + (2000 + 15u^2) = 0$$



Horizontal:

$$\ddot{x} = 0$$

$$\dot{x} = \int 0 \, dt$$

$$\dot{x} = c$$

$$\dot{x} = u \cos \theta$$

$$x = \int u \cos \theta \, dt$$

$$x = ut \cos \theta + c$$

$$x = ut \cos \theta$$

When  $x = 20, y = 15$

$$20 = ut \cos \theta$$

$$t = \frac{20}{u \cos \theta}$$

$$15 = -5 \left( \frac{20}{u \cos \theta} \right)^2 + u \left( \frac{20}{u \cos \theta} \right) \sin \theta$$

$$15 = -\frac{2000}{u^2 \cos^2 \theta} + \frac{20 \sin \theta}{\cos \theta}$$

$$15u^2 + 2000 \sec^2 \theta + 20u^2 \tan \theta = 0$$

$$15u^2 + 2000(1 + \tan^2 \theta) + 20u^2 \tan \theta = 0$$

$$15u^2 + 2000 + 2000 \tan^2 \theta + 20u^2 \tan \theta = 0$$

$$2000 \tan^2 \theta + 20u^2 \tan \theta + (2000 + 15u^2) = 0$$

Vertical:

$$\ddot{y} = -10$$

$$\dot{y} = \int -10 \, dt$$

$$\dot{y} = -10t + c$$

$$\dot{y} = -10t + u \sin \theta$$

$$y = \int -10t + u \sin \theta \, dt$$

$$y = -5t^2 + ut \sin \theta + c$$

$$y = -5t^2 + ut \sin \theta$$

| Marks | Criteria                                                                                                                                                                                                                                                       |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | Correct solution with logical process.                                                                                                                                                                                                                         |
| 1     | Finds the correct equations for displacement of both horizontal and vertical components having evaluated all constants correctly.<br>Or minor errors in equations but substitutes in and changes $\frac{1}{\cos^2 \theta} = \sec^2 \theta = 1 + \tan^2 \theta$ |

- ii. By considering the discriminant of this quadratic equation, find the minimum possible value of  $u$ .

2

$$\Delta = (20u^2)^2 - 4(2000)(2000 + 15u^2)$$

$$= 400u^4 - 120\,000u^2 - 16\,000\,000$$

$$= 400(u^4 - 300u^2 - 40\,000)$$

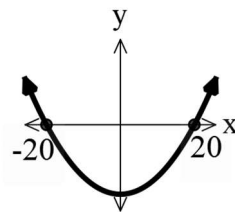
There must be at least 1 solution for  $\tan \theta \therefore \Delta \geq 0$

$$400(u^4 - 300u^2 - 40\,000) \geq 0$$

$$(u^2 - 400)(u^2 + 100) \geq 0$$

$$u^2 - 400 \geq 0 \quad (\text{since } u^2 + 100 > 0 \text{ for all } u)$$

$$(u - 20)(u + 20) \geq 0$$



$$u \leq -20 \text{ or } u \geq 20$$

But  $u$  is speed so  $u \geq 0$

$$\therefore \text{min value of } u = 20$$

| Marks | Criteria                                                                                                                 |
|-------|--------------------------------------------------------------------------------------------------------------------------|
| 2     | Solves the discriminant equation, finding the one correct value for $u$                                                  |
| 1     | Calculates the discriminant (correctly expanding it into quadratic form) and sets its value to $\geq 0$ , $> 0$ or $= 0$ |

- iii. Find the corresponding launch angle, correct to the nearest degree.

1

$$2000 \tan^2 \theta - 20 \times 20^2 \times \tan \theta + (2000 + 15 \times 20^2) = 0$$

$$2000 \tan^2 \theta - 8000 \tan \theta + 8000 = 0$$

$$\tan^2 \theta - 4 \tan \theta + 4 = 0$$

$$\tan \theta = \frac{4 \pm \sqrt{16 - 16}}{2}$$

$$\tan \theta = 2$$

$$\theta = 63.4^\circ$$

$$\theta = 63^\circ$$

| Marks | Criteria                                                                                                     |
|-------|--------------------------------------------------------------------------------------------------------------|
| 1     | Correct launch angle from found value of $u$ in part ii. Value of $u$ found cannot make the question easier. |

**End of Question 14**

**End of paper**